Will the Iran war undermine AI investment?

- **The Iran war could take some of the momentum out of the AI investment boom but we doubt that it will fundamentally disrupt the overall build-out.**
- The Iran war threatens AI investment in four main ways. **First, higher energy prices will raise the cost of running AI infrastructure, reducing projected returns.** Training and running AI models is notoriously energy-intensive, with energy estimated to account for between 15% and 25% of data centre operating costs. **Second, higher energy prices will also push up the cost of producing hardware for data centres, which could likewise weigh on returns.** The big exporters of AI-related equipment, Korea and Taiwan, are significant net energy importers and therefore particularly vulnerable to higher prices. (See here.)
- **Third, a global helium shortage could lead to a shortfall of advanced semiconductors.** About one third of global helium output is produced by Qatar. Helium's unique cooling properties make it difficult to replace with other materials and, because it is hard to store, firms tend to have only limited reserves. **Finally, recent events have also led to tighter financial conditions, both due to higher interest rates and weaker equity prices.** This could raise the costs of financing AI CapEx and risks firms dialling back their investment plans.
- **It therefore seems likely that global AI investment will take at least some hit from the effects of the war, but in our baseline scenario this should be limited.** (See here for more on our Iran war scenarios.) Most AI investment takes place in the US where natural gas is the main source of electricity for data centres. (See here.) Thanks to abundant domestic supplies, US natural gas prices have been little changed in the past month. While they are likely to rise in the coming years, we expect them to remain far lower than elsewhere.
- Meanwhile, global chipmakers will prioritise higher-value output and are also likely to prove relatively insensitive to the price needed to secure their share of reduced global helium supply. This should shield hyperscalers from prolonged supply disruptions, albeit not higher prices. To the extent that costs rise, AI providers may assume that demand for their products will be fairly price inelastic due to the potential productivity-enhancing benefits (see here), so they may be comfortable in assuming that they can pass on those costs rather than accept lower returns. Finally, most AI CapEx is funded out of the hyperscalers' cashflows, rather than borrowing, meaning that slightly higher interest rates should have limited impact. (See here.)
- **Given the sharp projected rise in AI investment in the next few years, a modest reduction in spending – particularly a short-lived one – would still result in strong growth.** Analysts' projections suggest that the big five US hyperscalers will raise CapEx from $425bn last year to over $700bn in 2028. (See Chart 1.)
- However, **the risk of prolonged disruption would be greater in our adverse scenario, in which global energy prices rise much further.** We assume US natural gas prices would follow a similar trajectory in that scenario, but there would be a greater chance of AI hardware prices surging – or hardware being unavailable – due to energy or helium supply disruptions. That would come at a time when there is evidence that hardware costs are already rising sharply. (See Chart 2.) Given the negative market reaction to some big tech firms' CapEx plans following earnings calls earlier this year (see here), even the hyperscalers might be forced to put those plans on ice if a further surge in costs coincided with more meaningful weakness in global equity markets, which would be a distinct risk in our adverse scenario.
- **Nonetheless, the ability of firms to adapt leads us to think that the longer-term outlook for AI remains robust.** Higher energy prices will incentivise AI firms to invest more in their own power generation and storage, as well as to improve the energy efficiency of their AI models. Google's recent announcement of

Vicky Redwood, Chief Economic Adviser, victoria.redwood@capitaleconomics.com
Stephen Brown, Chief North America Economist, stephen.brown@capitaleconomics.com

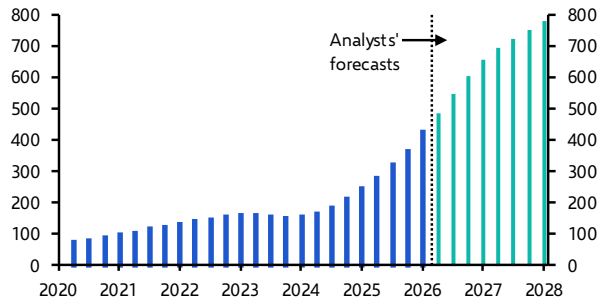
Learn more: Consultancy Events Data

a new algorithm to drastically reduce memory requirements shows the incentives that much higher costs can have in spurring innovation. Due to recent events, chipmakers will also be incentivised to invest more in advanced systems to recycle helium and further diversify their supply chains of critical inputs.

- **Even if the Iran war does not have major consequences for the overall levels of AI investment in the long run, it could have lasting effects on the *location* of investment.** The Middle East has been a popular location for data centres due to its abundance of cheap energy; that now seems likely to change given increased security concerns. Meanwhile, other areas with cheap energy are likely to look even more attractive than before. This includes parts of the US but also some countries outside the US such as the Nordics.
- **While most of our immediate concerns about AI investment relate to supply disruptions, a bigger concern for our long-run optimistic view on AI would be if the fallout from the Iran war were to seriously slow the adoption of AI.** That could happen if, for example, AI model providers were forced to hike prices at the same time as firms in many economies were suffering from much higher energy bills. That said, if the potential productivity gains from AI are as large as we expect, modest price rises are unlikely to materially change firms' incentives to adopt AI. And because new technologies tend to decline in cost as they mature, any Iran-related price rises are likely to be outweighed over time by these structural cost reductions. Finally, while higher energy prices may affect the ability of smaller firms with limited budgets to adopt AI, history shows that recessions also provide a strong incentive for firms more generally to speed up the adoption of labour-saving technologies. Accordingly, a global recession could, counterintuitively, accelerate AI adoption in some instances.

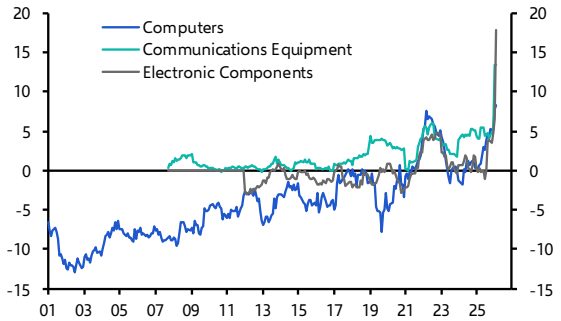
All our work on AI, including our original 2023 Spotlight report "AI, Economies & Markets", is available here.)

Chart 1: Combined CapEx of 5 US Hyperscalers ($bn)



Sources: I/B/E/S forecasts via LSEG, Capital Economics

Chart 2: US Producer Price Indices (% y/y)



Sources: LSEG, Capital Economics

Disclaimer: While every effort has been made to ensure that the data quoted and used for the research behind this document is reliable, there is no guarantee that it is correct, and Capital Economics Limited and its subsidiaries can accept no liability whatsoever in respect of any errors or omissions. This document is a piece of economic research and is not intended to constitute investment advice, nor to solicit dealing in securities or investments.

Distribution: Subscribers are free to make copies of our publications for their own use, and for the use of members of the subscribing team at their business location. No other form of copying or distribution of our publications is permitted without our explicit permission. This includes but is not limited to internal distribution to non-subscribing employees or teams.